COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/22/26	**DATE SUBMITTED:**	10/22/26
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ACTIVITY NO. 12.1 **Build a Sample Web App in a Docker Container**

1. PROCEDURE OUTPUT

Part 1: Launch the DEVASC VM

If you have not already completed the Lab - Install the Virtual Machine Lab Environment, do so now. If you have already completed that lab, launch the DEVASC VM now.



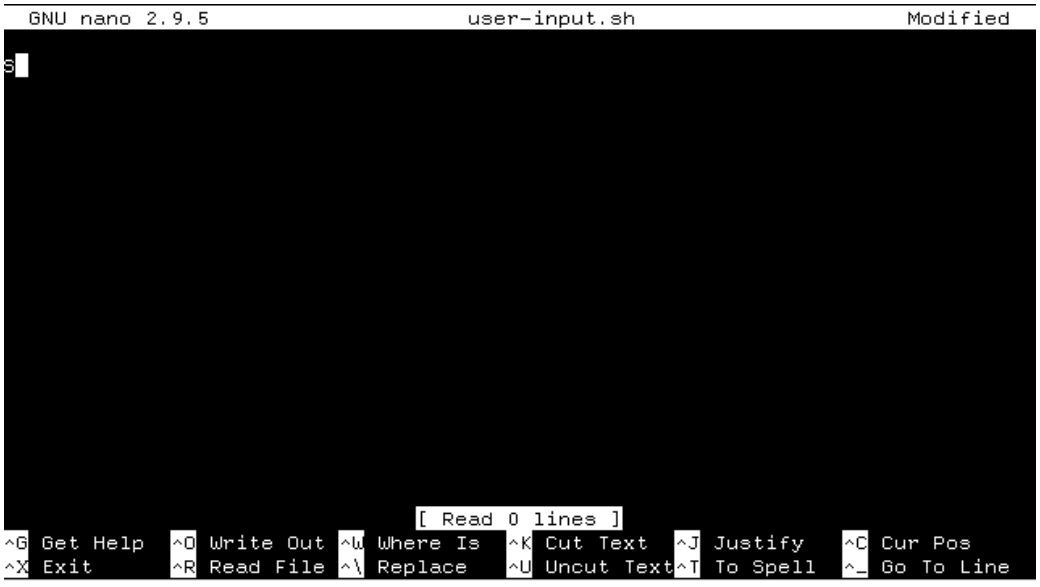
Part 2: Create an empty bash script file.

Bash knowledge is crucial for working with continuous integration, continuous deployment, containers, and with your development environment. Bash scripts help programmers automate a variety of tasks in one script file. In this part, you will briefly review how to create a bash script. Later in the lab, you will use a bash script to automate the creation of a web app inside of a Docker container.

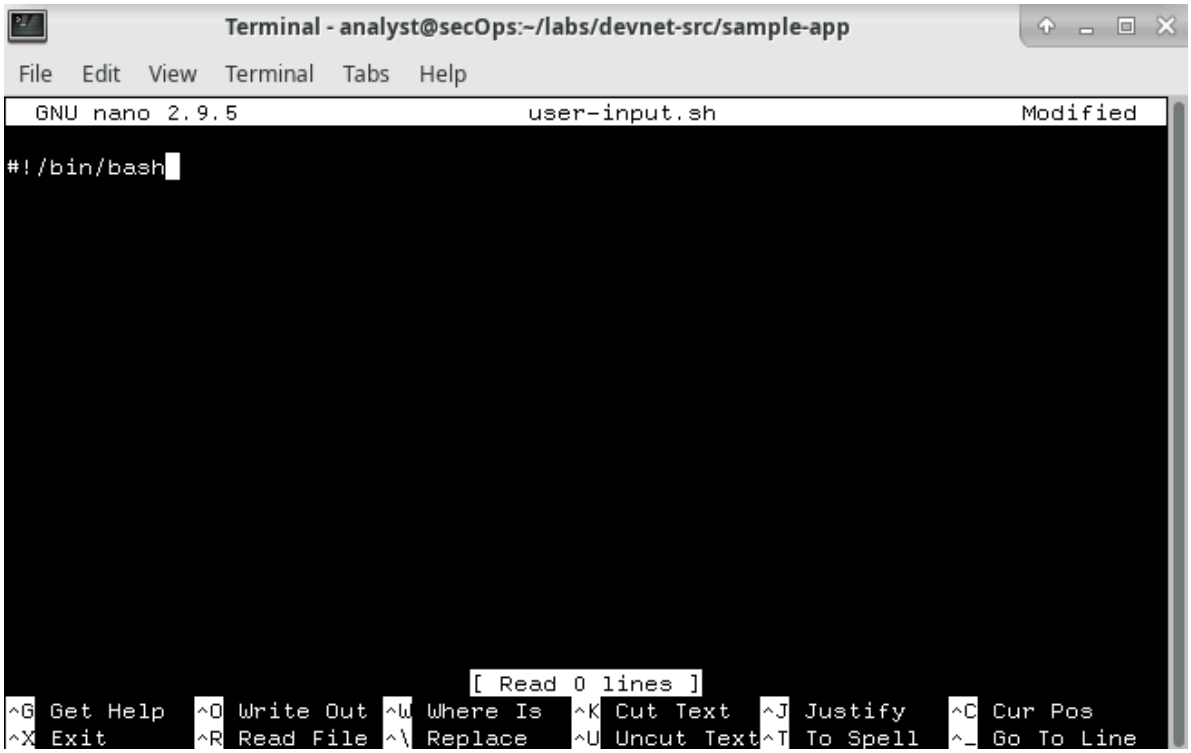
Step 1:

```
[analyst@secOps ~]$ mkdir labs/devnet-src/sample-app/
[analyst@secOps ~]$ cd labs/devnet-src/sample-app/
```

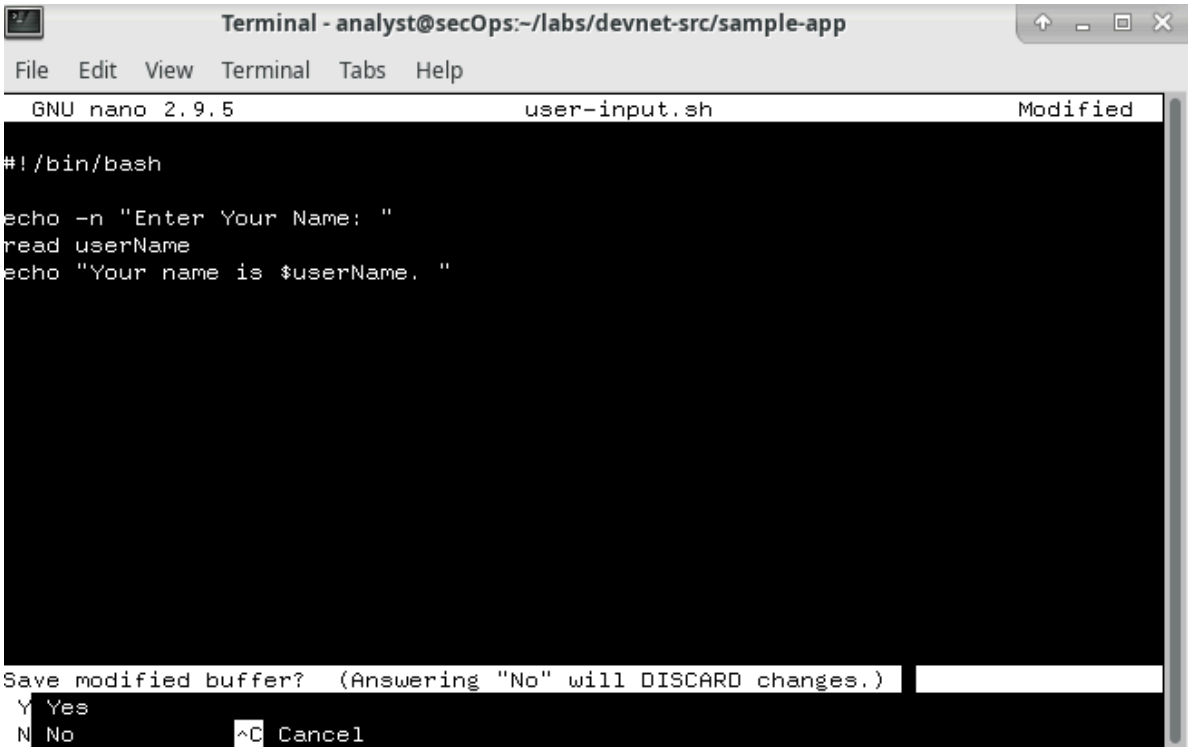
Step 2:



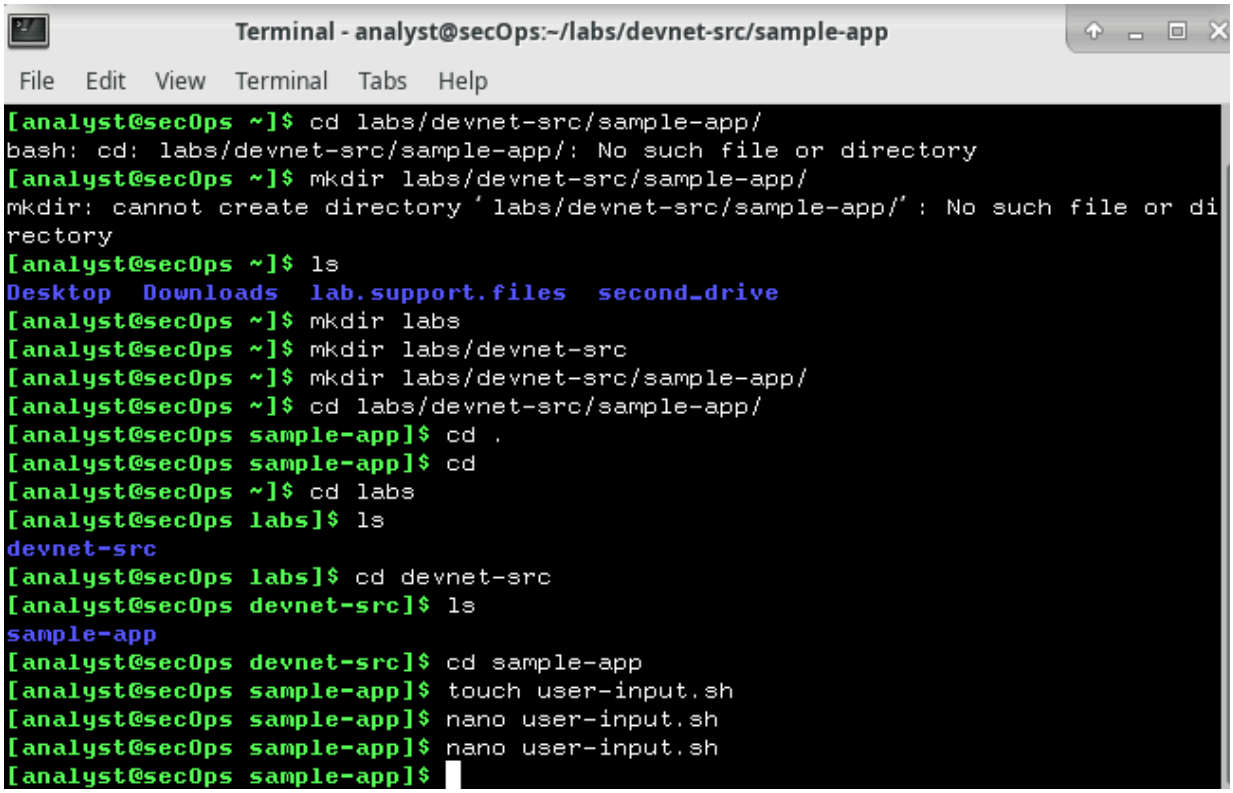
Step 3:


Terminal - analyst@secOps:~/labs/devnet-src/sample-app
File Edit View Terminal Tabs Help
GNU nano 2.9.5 user-input.sh Modified
#!/bin/bash
[Read 0 lines]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line

Step 4:


Terminal - analyst@secOps:~/labs/devnet-src/sample-app
File Edit View Terminal Tabs Help
GNU nano 2.9.5 user-input.sh Modified
#!/bin/bash
echo -n "Enter Your Name: "
read userName
echo "Your name is $userName. "
Save modified buffer? (Answering "No" will DISCARD changes.)
Y Yes
N No ^C Cancel

Step 5:



Step 6:

```
[analyst@secOps sample-app]$ bash user-input.sh
Enter Your Name: Vanderohe
Your name is Vanderohe.
[analyst@secOps sample-app]$
```

Step 7:

```
[analyst@secOps sample-app]$ ls -l user-input.sh
-rw-r--r-- 1 analyst analyst 87 Oct 21 20:18 user-input.sh
[analyst@secOps sample-app]$ chmod a+x user-input.sh
[analyst@secOps sample-app]$ ls -l user-input.sh
-rwxr-xr-x 1 analyst analyst 87 Oct 21 20:18 user-input.sh
```

Step 8:

```
[analyst@secOps sample-app]$ mv user-input.sh user-input
```

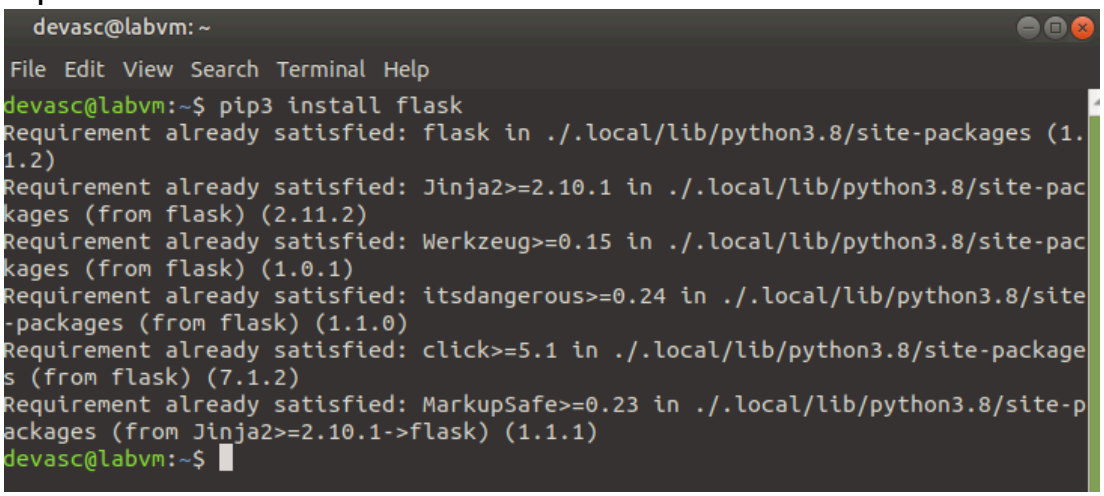
Step 9:

```
[analyst@secOps sample-app]$ ./user-input
Enter Your Name: Vanderohe
Your name is Vanderohe.
```

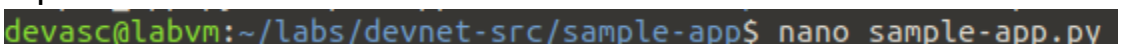
Part 3: Create a Sample Web App

Before we can launch an application in a Docker container, we first need to have the app. In this part, you will create a very simple Python script that will display the IP address of the client when the client visits the web page.

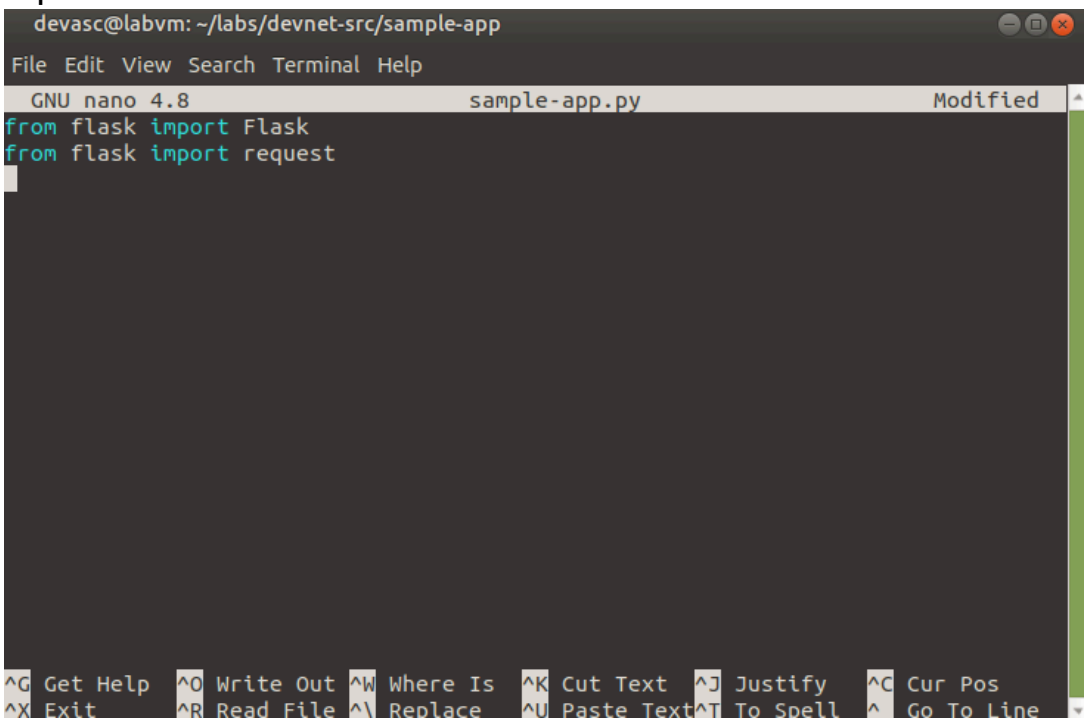
Step 1:



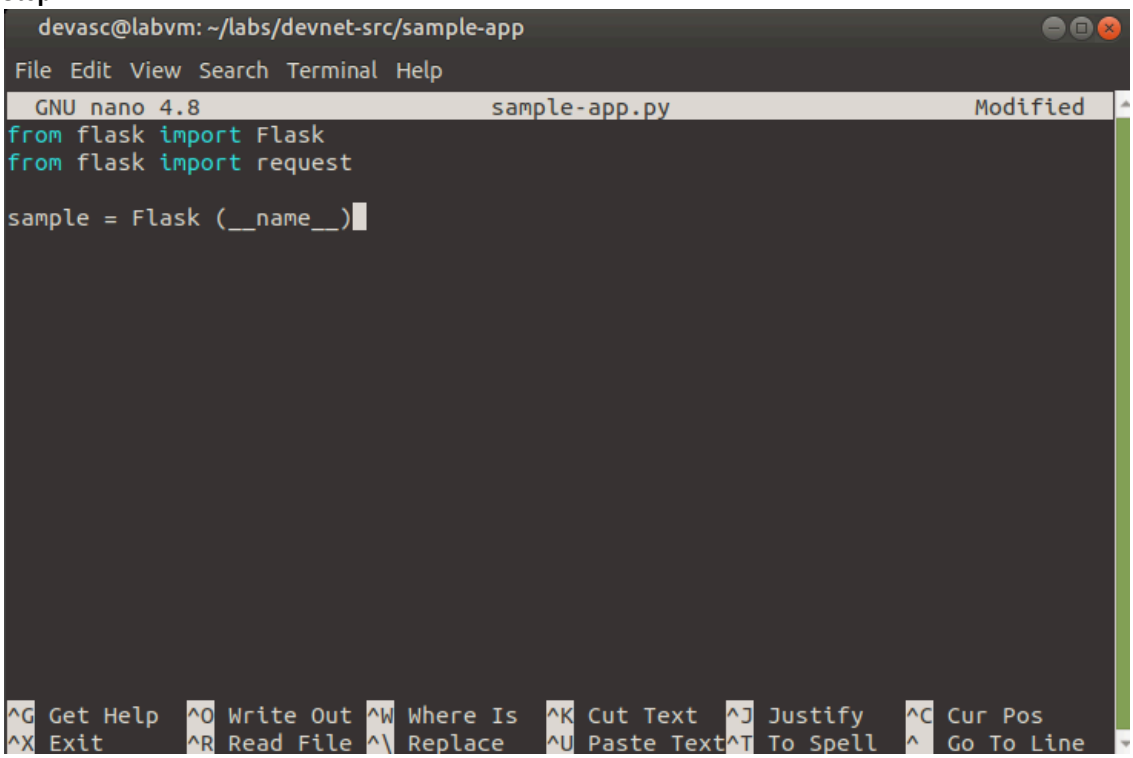
Step 2:



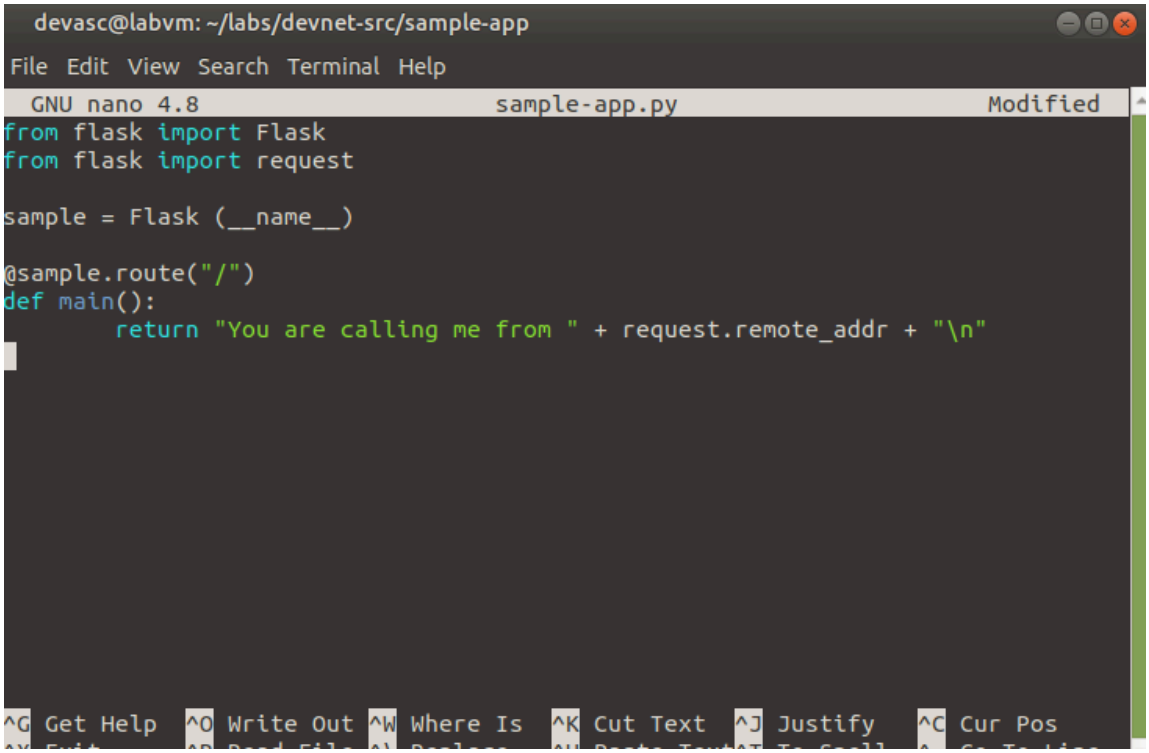
Step 3:



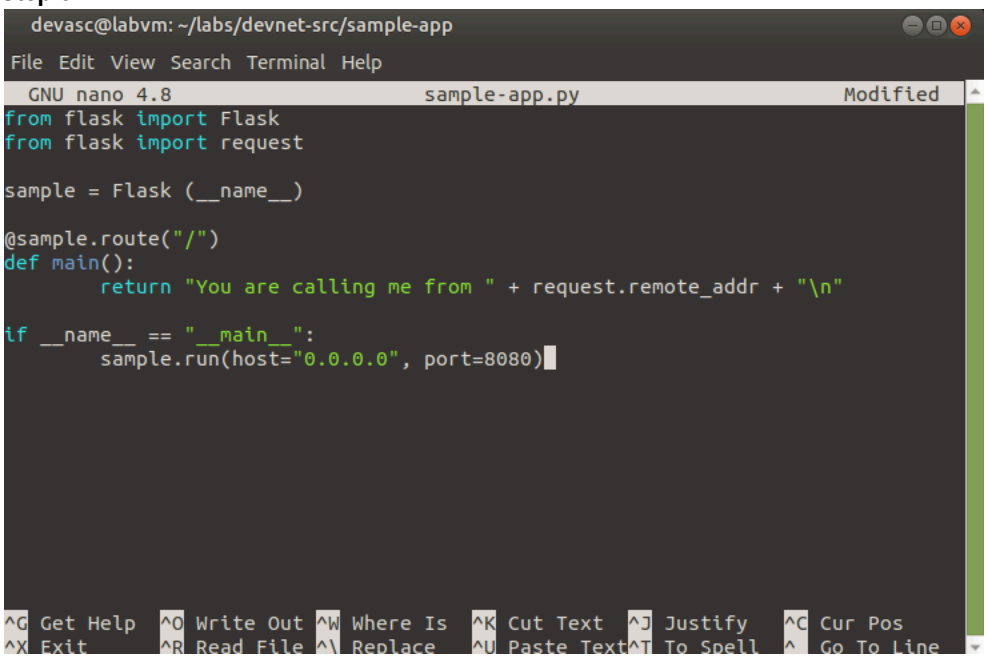
Step 4:


devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.py Modified
from flask import Flask
from flask import request
sample = Flask (__name__)
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

Step 5:


devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.py Modified
from flask import Flask
from flask import request
sample = Flask (__name__)
@sample.route("/")
def main():
return "You are calling me from " + request.remote_addr + "\n"
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos

Step 6:



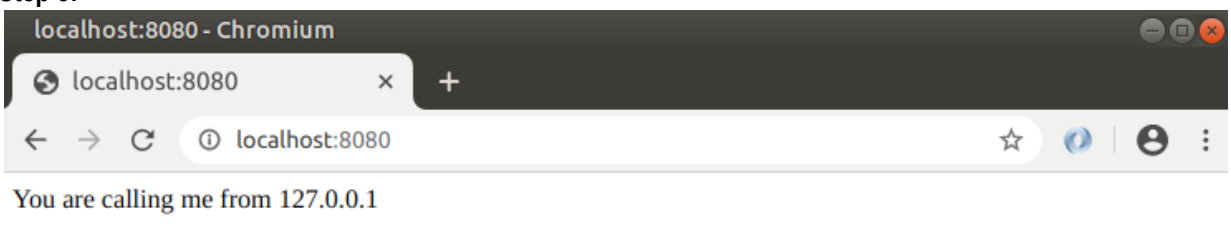
Step 7:

```
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py


Use "fg" to return to nano.

[1]+  Stopped                 nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
 * Serving Flask app "sample_app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployme
nt.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
```

Step 8:



```
devasc@labvm: ~
File  Edit  View  Search  Terminal  Help
devasc@labvm:~$ curl http://0.0.0.0:8080
You are calling me from 127.0.0.1
devasc@labvm:~$
```

Step 9:

```
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
 * Serving Flask app "sample_app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deploym
nt.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [22/Oct/2026 07:08:23] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [22/Oct/2026 07:08:23] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [22/Oct/2026 07:09:37] "GET / HTTP/1.1" 200 -
^Cdevasc@labvm:~/labs/devnet-src/sample-app$
```

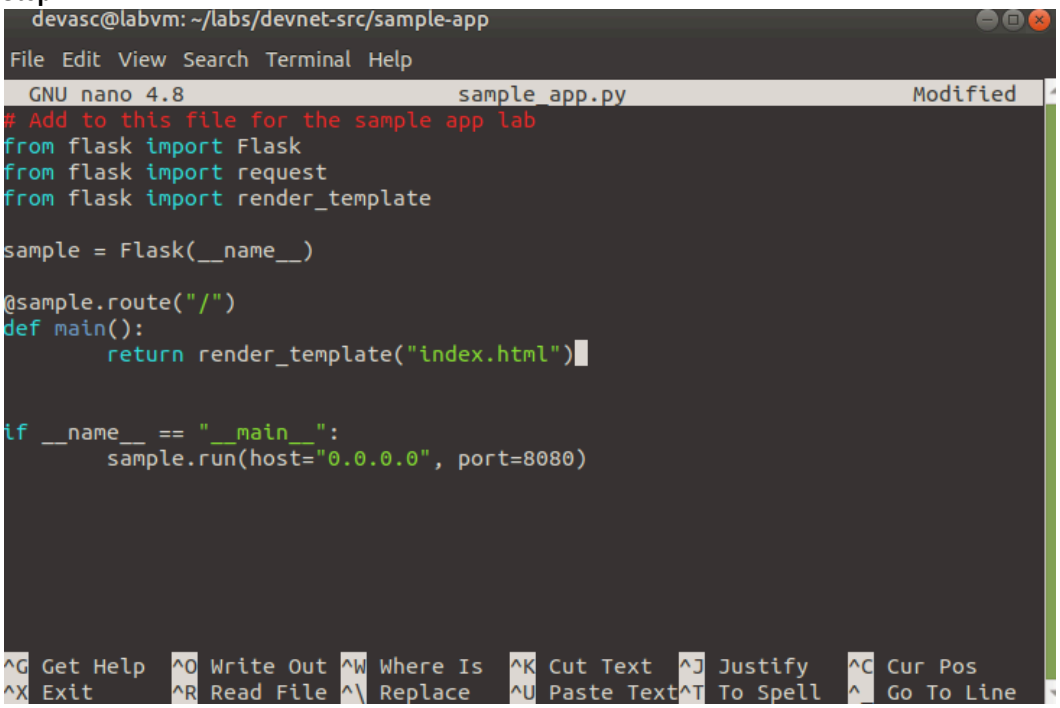
Part 4: Configure the Web App to use Website Files

In this part, build out the sample web app to include an index.html page and style.css specification. The index.html is typically the first page loaded in a client's web browser when visiting your website. The style.css is a style sheet used to customize the look of the web page.

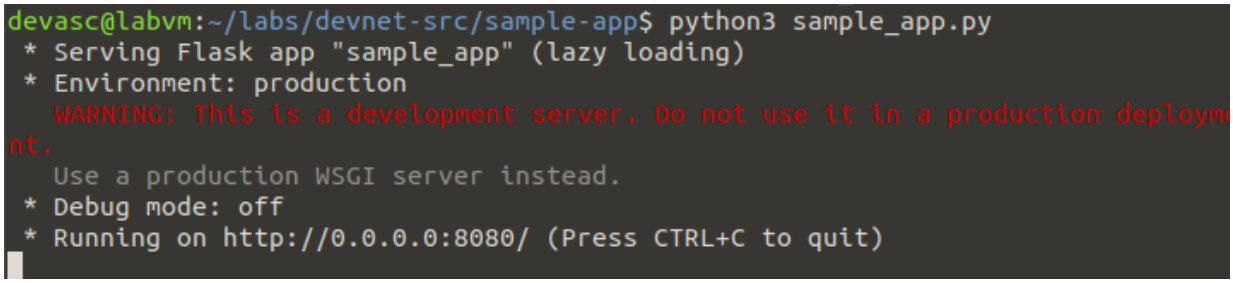
Step 1:

```
devasc@labvm:~/labs/devnet-src/sample-app$ cat templates/index.html
<html>
<head>
    <title>Sample app</title>
    <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
    <h1>You are calling me from {{request.remote_addr}}</h1>
</body>
</html>
devasc@labvm:~/labs/devnet-src/sample-app$ cat static/style.css
body {background: lightsteelblue;}
```

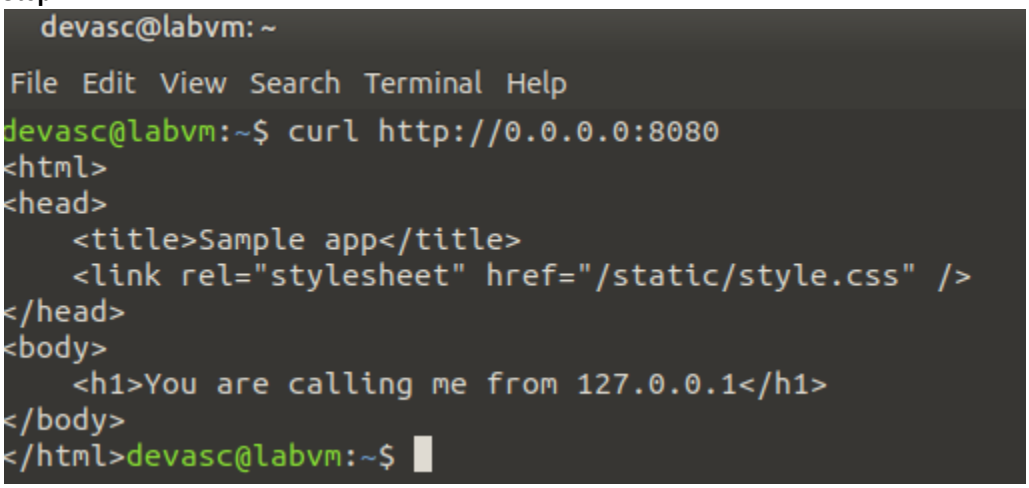
Step 2:



Step 3:



Step 4:



Step 5:

```
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
 * Serving Flask app "sample_app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deploym
nt.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
127.0.0.1 - - [22/Oct/2026 07:23:11] "GET / HTTP/1.1" 200 -
^Cdevasc@labvm:~/labs/devnet-src/sample-app$
```

Part 5: Create a Bash Script to Build and Run a Docker Container

An application can be deployed on a bare metal server (physical server dedicated to a single-tenant environment) or in a virtual machine, like you just did in the previous Part. It can also be deployed in a containerized solution like Docker. In this part, you will create a bash script and add commands to it that complete the following tasks to build and run a Docker container:

- Create temporary directories to store the website files.
- Copy the website directories and sample_app.py to the temporary directory.
- Build a Dockerfile.
- Build the Docker container.
- Start the container and verify it is running.

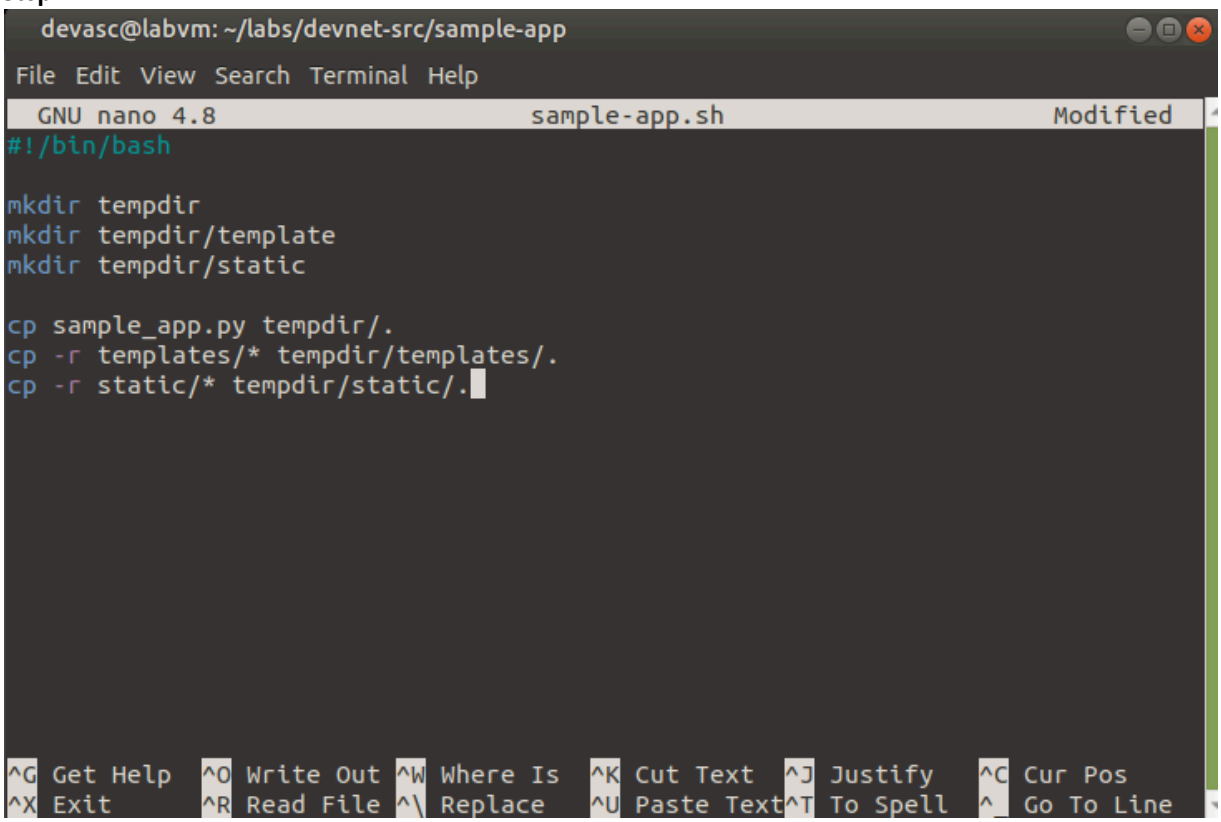
Step 1:

```
devasc@labvm: ~/labs/devnet-src/sample-app
File  Edit  View  Search  Terminal  Help
  GNU nano 4.8                    sample-app.sh                         Modified
#!/bin/bash

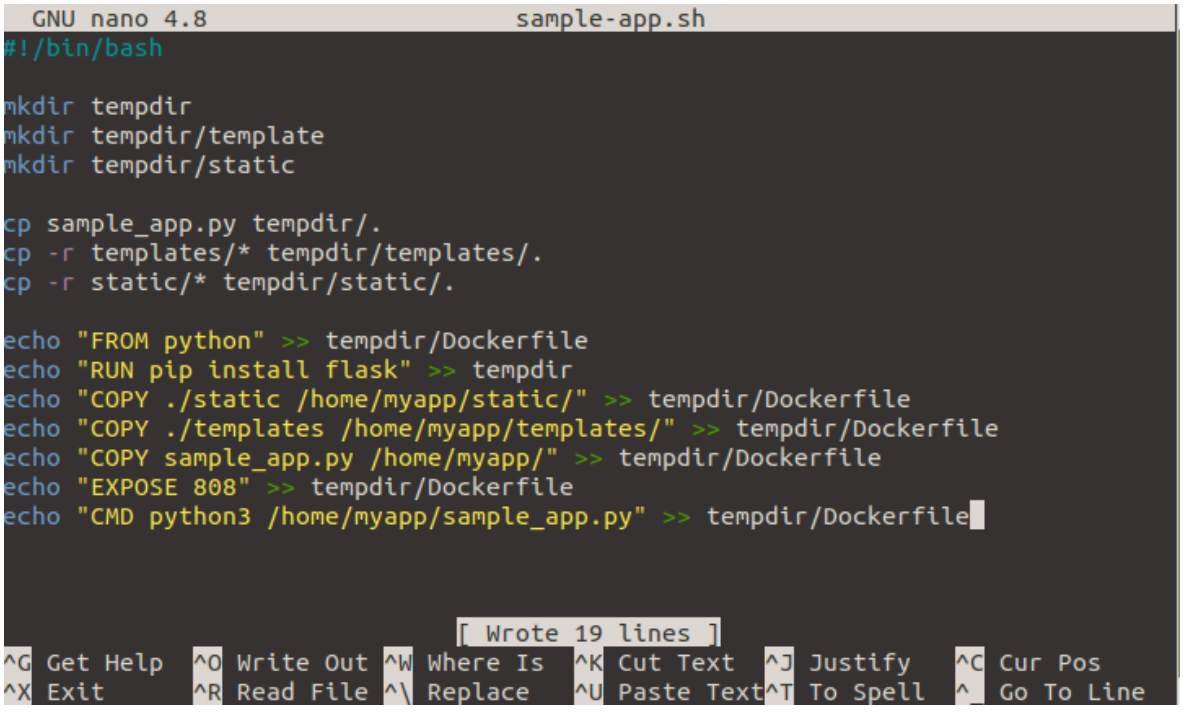
mkdir tempdir
mkdir tempdir/template
mkdir tempdir/static

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File  ^\ Replace   ^U Paste Text^T To Spell  ^_ Go To Line
```

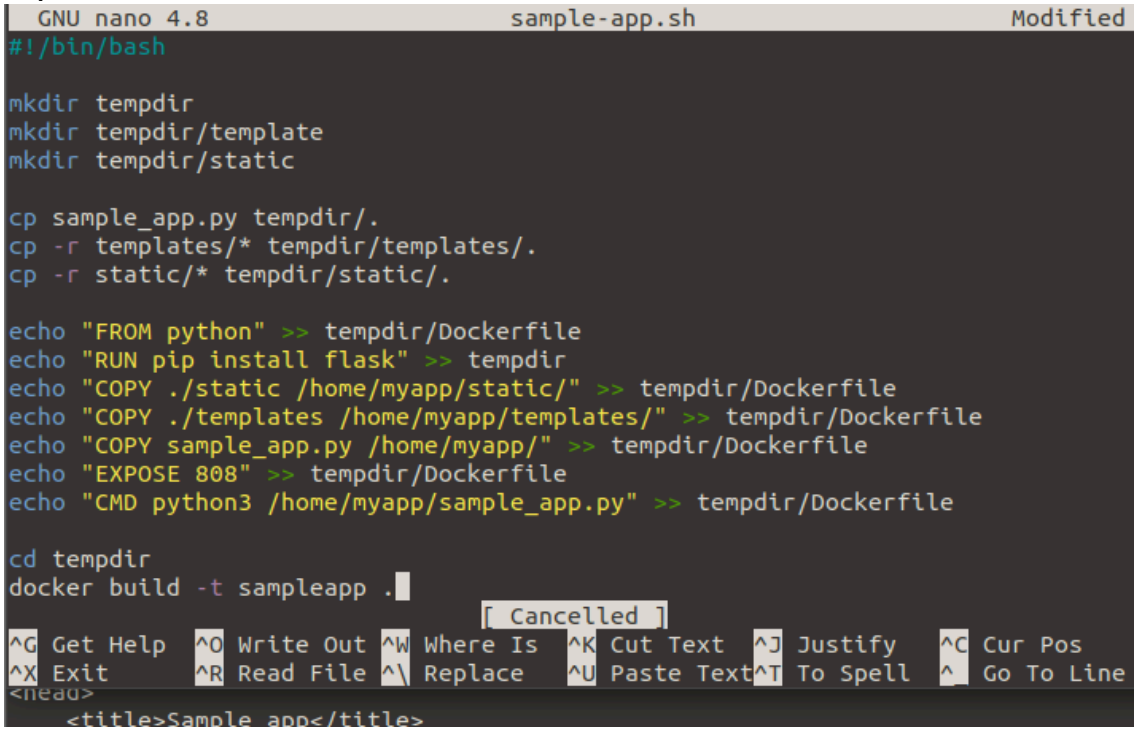
Step 2:


devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.sh Modified
#!/bin/bash
mkdir tempdir
mkdir tempdir/template
mkdir tempdir/static
cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

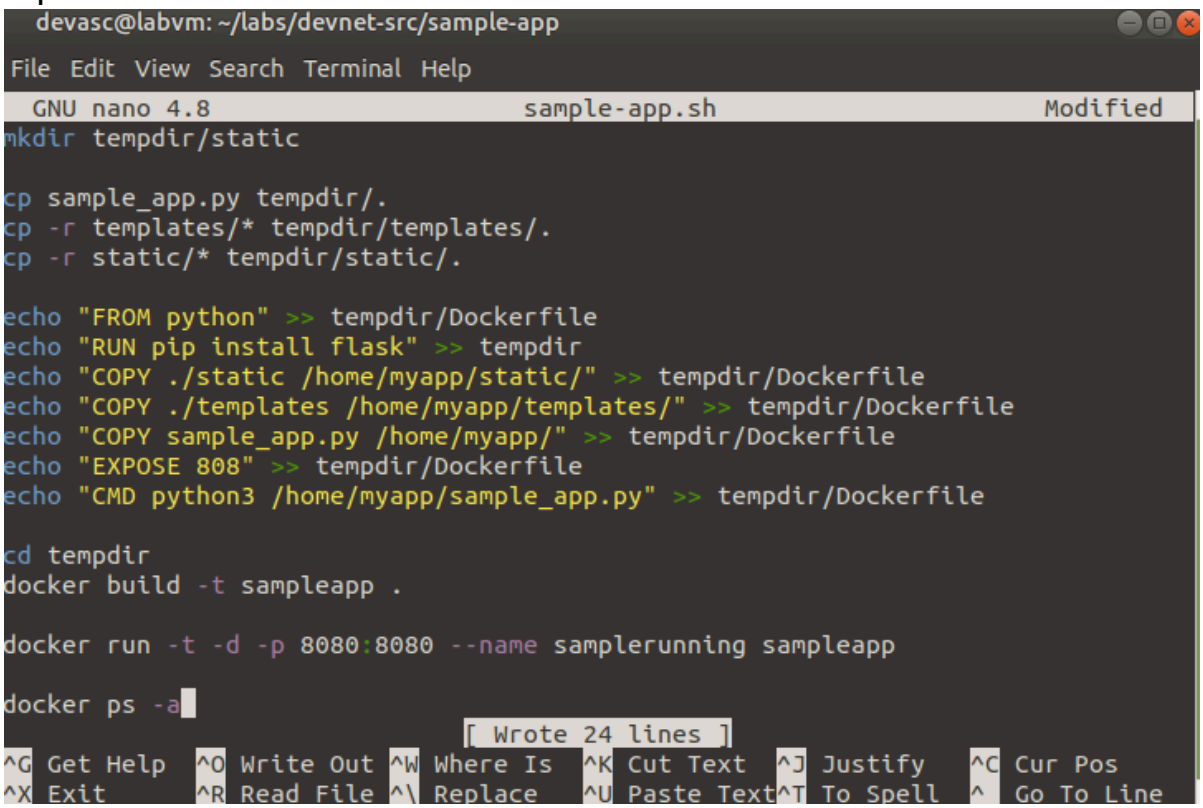
Step 3:


GNU nano 4.8 sample-app.sh
#!/bin/bash
mkdir tempdir
mkdir tempdir/template
mkdir tempdir/static
cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.
echo "FROM python" >> tempdir/Dockerfile
echo "RUN pip install flask" >> tempdir
echo "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
echo "EXPOSE 808" >> tempdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile
[Wrote 19 lines]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

Step 4:


GNU nano 4.8 sample-app.sh Modified
#!/bin/bash
mkdir tempdir
mkdir tempdir/template
mkdir tempdir/static
cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.
echo "FROM python" >> tempdir/Dockerfile
echo "RUN pip install flask" >> tempdir
echo "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
echo "EXPOSE 808" >> tempdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile
cd tempdir
docker build -t sampleapp .
[Cancelled]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line
<head>
<title>Sample app</title>

Step 5:


devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.sh Modified
mkdir tempdir/static
cp sample_app.py tempdir/.
cp -r templates/* tempdir/templates/.
cp -r static/* tempdir/static/.
echo "FROM python" >> tempdir/Dockerfile
echo "RUN pip install flask" >> tempdir
echo "COPY ./static /home/myapp/static/" >> tempdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tempdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tempdir/Dockerfile
echo "EXPOSE 808" >> tempdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tempdir/Dockerfile
cd tempdir
docker build -t sampleapp .
docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a
[Wrote 24 lines]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line

Step 6:

```
                                [ Wrote 24 lines ]
^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Paste Text^T To Spell  ^_ Go To Line
```

Part 6: Build, Run, and Verify the Docker Container

In this part, you will execute bash script which will make the directories, copy over the files, create a Dockerfile, build the Docker container, run an instance of the Docker container, and display output from the docker ps -a command showing details of the container currently running. Then you will investigate the Docker container, stop the container from running, and remove the container.

Note: Be sure you stopped any other web server processes you may still have running from the previous parts of this lab.

Step 1:

```
devasc@labvm:~/labs/devnet-src/sample-app$ bash ./sample-app.sh
Sending build context to Docker daemon  6.144kB
Step 1/7 : FROM python
latest: Pulling from library/python
8d0037029946: Pull complete
63040bd9752a: Pull complete
7046f4e749f0: Pull complete
992d9e06c2e8: Pull complete
6f634d75d986: Pull complete
23346f0ec4f3: Pull complete
3c16dcb966e6: Pull complete
```

Step 2:

```
devasc@labvm:~/labs/devnet-src/sample-app$ ls tempdir/
Dockerfile  sample_app.py  static  templates
devasc@labvm:~/labs/devnet-src/sample-app$ cat tempdir/Dockerfile
FROM python
RUN pip install flask
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
```

```
devasc@labvm:~/labs/devnet-src/sample-app$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
```

```
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DO
N group default
    link/ether 02:42:ba:f6:91:fa brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::42:baff:fef6:91fa/64 scope link
       valid_lft forever preferred_lft forever
```

```
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps -a
CONTAINER ID        IMAGE               COMMAND                  CREATED
     STATUS                      PORTS               NAMES
8524b8505b2c        967555bbc8c4        "/bin/sh -c 'pip ins…"   7 minutes ago
     Exited (2) 7 minutes ago                        zen_hellman
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Solution for errors encountered**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is solely used for problem solving of errors encountered during the lab activity**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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